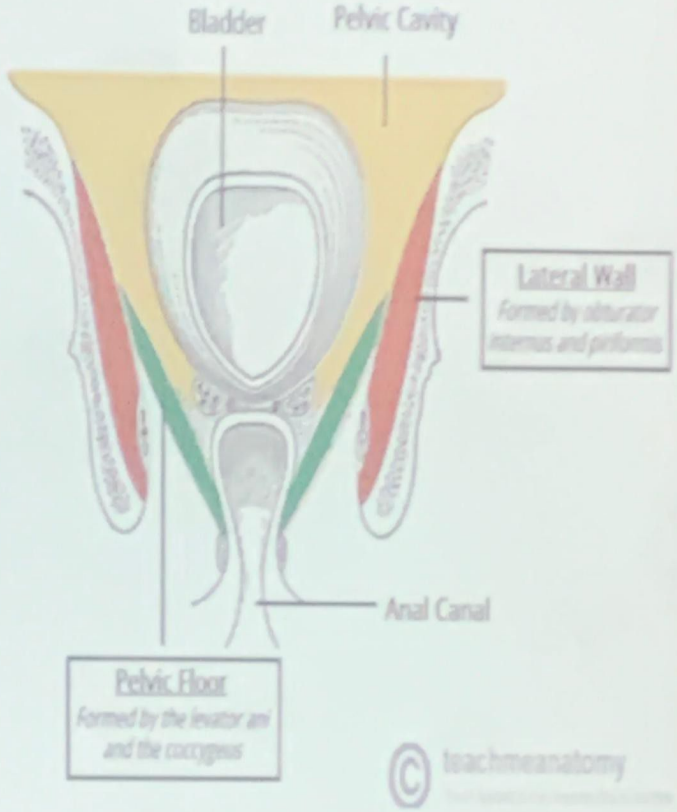
Pelvic Diaphragm

- Introduction
- Muscles of pelvic floor
- Some role of pelvic floor muscles
- Applied anatomy

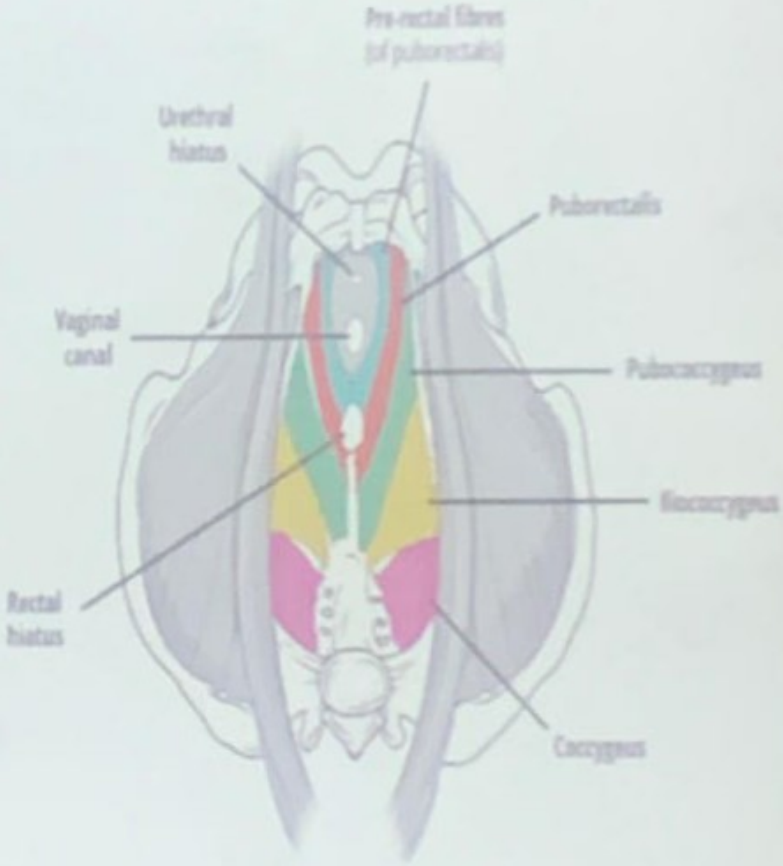
Pelvic floor: It is composed of levator ani and coccygeus muscles together with their fascia. The muscles form a sling across the pelvic cavity holding and supporting the pelvic viscera.

The muscles of the pelvic floor include coccygeus and levator ani. The levator ani is made up of iliococcygeus and pubococcygeus. These muscles arise in continuity from the spine of the ischium, from the white line over the obturator fascia and from the body of pubis and are inserted into the coccyx and anococcygeal ligament with the deepest part called anococcygeal raphe formed by interdigitating fibres of levator ani.



Coccygeus: It is also known as ischiococcygeus in man. In tailed animals is called agitor caudae because of the tail to wag. The coccygeus (or ischiococcygeus) is the smaller, and most posterior pelvic floor component. It's gluteal surface is fibrotic forming sacrospinous ligament. It originates from the tip of ischial spine along the posterior margin of obturator internus. The fibres fan out laterally and backward to insert at the lowest pieces of sacrum and coccyx which is edge to edge to piriformis.

Innervation: perineal branch of S_4 and S_5.



Levator Ani: It is consist of two muscles, iliococcygeus and pubococcygeus separated by a triangular space. Their fibres arise from the body of pubis across the obturator fascia extending to ischial spine. The obturator fascia forms a thickening, the white line. The muscle fibres is densely adherent to the fascia.

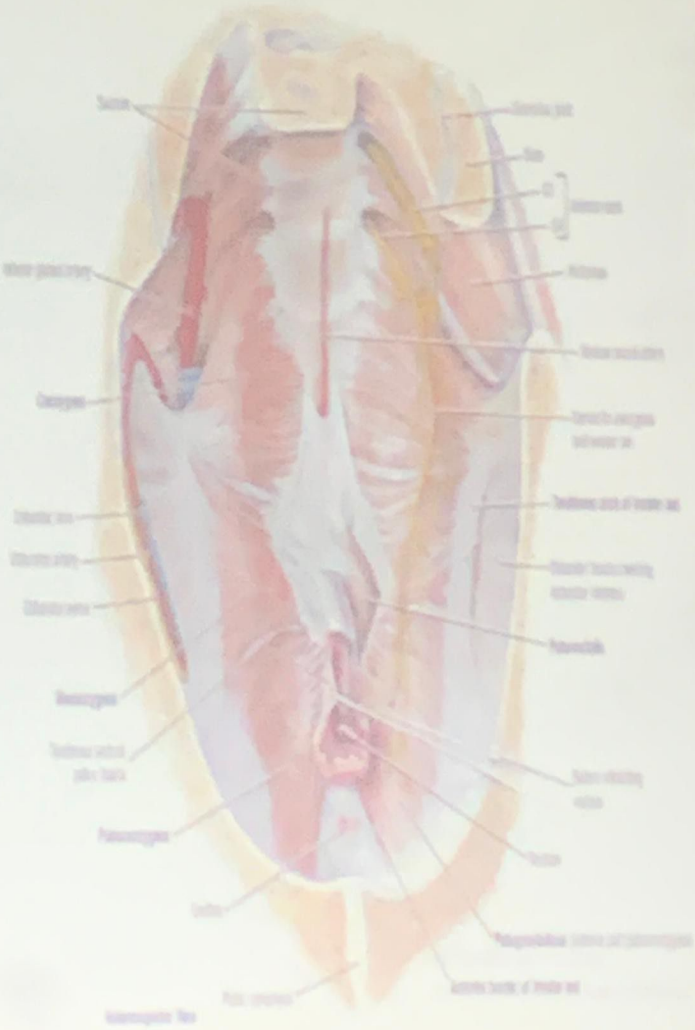
Iliococcygeus: It has thin muscle fibres. It arises from the posterior half of the white line and the pelvic surface of the ischial spine crossing the pelvic surface of coccygeus to the sides of coccyx and anococcygeal ligament and raphe.



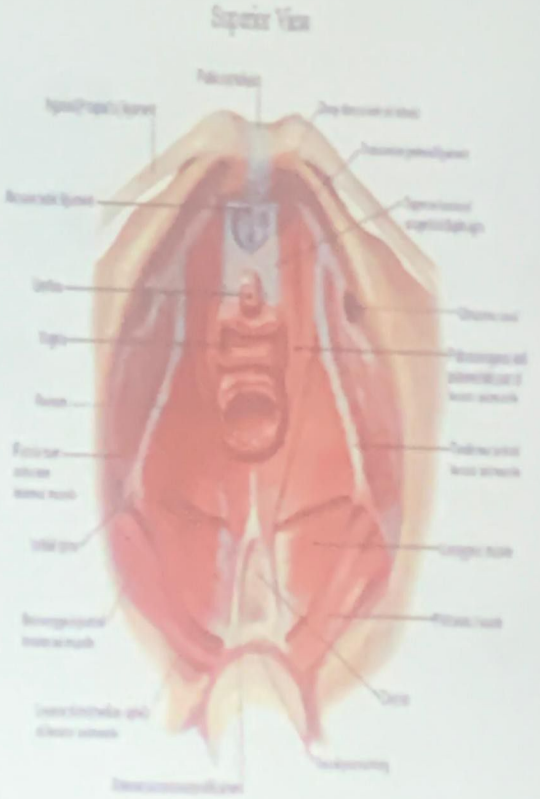
It extends from the tip of coccyx to anorectal junction.

This part of the levator ani is the actual "levator" of the three: its action elevates the pelvic floor and the anorectal canal.

Clinical point: During defaecation, and second stage of labour, it is passively stretched.

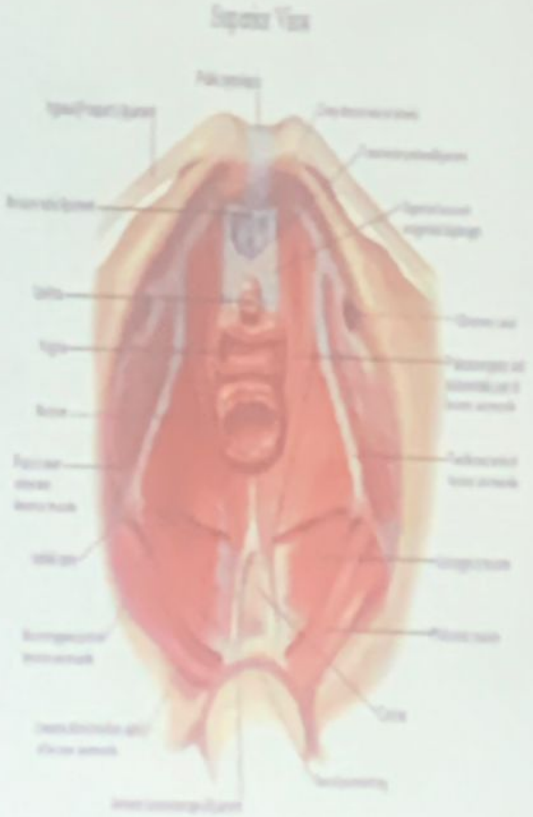


Pubococcygeus: It is the most important of the levator ani arising from the anterior half of the white line and from the posterior surface of the body of pubis and pubic symphysis. It forms flat sheath that sweeps backwards overlapping iliococcygeus. Bulk of the muscle is inserted at the tip of coccyx, anorectal ligament and raphe. It has the following components puborectalis, levator prostatae in males and pubovaginalis in females.



Puborectalis: Fibres arising more anteriorly from the periosteum of the body of pubis sweep more medially and more inferiorly around the anorectal junction and joining the opposite side with no raphe forming a U-shaped sling, extending from the bodies of the pubic bones, past the **urogenital hiatus**, around the anal canal known as puborectalis. Its tonic contraction bends the canal anteriorly, creating the anorectal angle (90 degrees) at the **anorectal junction** (where the rectum meets the anus).

The main function of this thick muscle is to maintain faecal continence – during defecation this muscle relaxes.



Some fibres of the puborectalis muscle (pre-rectal fibres) form another U-shaped sling that flank the urethra in the male known as levator prostatae and the urethra and vagina in the female called pubovaginalis or sphincter urethra / vaginae. These fibres are very important in preserving urinary continence, especially during abrupt increase of the intra-abdominal pressure i.e. during sneezing.

The roles of the pelvic floor muscles are:

Support of abdominopelvic viscera (bladder, intestines, uterus etc.) through their tonic contraction.

Resistance to increases in intra-pelvic/abdominal pressure during activities such as coughing or lifting heavy objects.

Urinary and faecal continence. The muscle fibres have a sphincter action on the rectum and urethra. They relax to allow urination and defecation.

Clinical relevance: Pelvic floor dysfunction refers to a range of signs and symptoms that relates to abnormal functioning of the pelvic floor muscles.

In women, the pelvic floor muscles support the urethra, vagina, and anal canal. The weakening of these muscles can result in a loss of structural support to these organs – presenting as:

- Urinary incontinence
- Faecal incontinence
- Genitourinary prolapse
- Pelvic pain
- Sexual dysfunction

The causes of pelvic floor dysfunction are understood to be **multifactorial** and include obstetric trauma, increasing age, obesity, and chronic straining.

